

PORTGLENONE, United Kingdom

WMO: 039150

Lat: 54.8653N

Lon: 6.4583W

Elev: 65

StdP: 100.55

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	-1.0	-4.9	2.5	0.1	-3.6	2.8	1.1	17.9	5.1	13.6	6.7	1.3	130	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	22.4	17.3	20.6	16.5	19.1	15.7	18.2	21.1	17.2	19.6	16.3	18.4	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.2	19.3	16.1	11.5	18.4	15.3	10.9	17.5	51.5	21.2	48.5	19.6	46.0	18.6	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.3	9.6	8.4	-5.3	24.8	2.2	1.6	-6.9	26.0	-8.2	26.9	-9.5	27.8	-11.1	29.0
			-5.5	19.8	2.1	1.2	-7.1	20.7	-8.3	21.4	-9.5	22.1	-11.0	23.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	4.7	5.0	6.1	8.1	10.6	13.1	14.7	14.6	12.8	10.1	7.0	5.0
	DBStd	4.46	2.74	2.59	2.58	2.58	2.67	2.40	2.26	1.99	2.08	2.44	2.71	3.18
	HDD10.0	806	166	142	122	68	24	3	0	0	2	29	95	156
	HDD18.3	3289	423	374	378	307	240	159	115	117	167	256	341	413
	CDD10.0	563	1	1	2	11	42	94	146	143	86	32	4	1
	CDD18.3	5	0	0	0	0	0	1	3	2	0	0	0	0
	CDH23.3	26	0	0	0	0	0	8	12	6	0	0	0	0
	CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0
(14) Wind	WSAvg	3.8	4.5	4.2	4.1	3.9	3.6	3.4	3.2	3.3	3.5	4.0	4.0	4.2
(15) Precipitation	PrecAvg	1072	105	80	76	74	73	75	85	90	80	116	112	108
	PrecMax	1323	175	147	157	126	169	164	136	223	186	218	182	223
	PrecMin	846	29	29	20	30	12	30	38	15	1	45	60	53
	PrecStd	115	35	31	32	28	37	33	26	45	40	50	36	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	12.7	14.5	18.2	22.2	24.1	25.0	24.1	20.4	16.7	14.2	13.0
		MCWB	10.8	10.9	10.7	12.8	16.3	17.2	19.4	18.5	17.3	14.6	13.2	11.9
	2%	DB	10.9	11.3	12.4	15.6	19.3	21.2	22.6	21.2	18.8	15.5	13.0	11.8
		MCWB	9.9	9.9	9.9	11.4	14.6	16.3	17.7	17.2	16.0	14.0	12.0	10.9
	5%	DB	9.9	10.2	11.3	13.7	17.1	19.2	20.6	19.4	17.4	14.7	12.0	10.8
		MCWB	9.0	9.0	9.2	10.4	13.3	15.2	16.8	16.2	15.0	13.2	11.1	9.9
	10%	DB	8.9	9.1	10.3	12.4	15.3	17.5	18.9	18.2	16.3	13.9	11.1	9.7
		MCWB	8.1	8.0	8.4	9.6	12.1	14.1	15.9	15.4	14.1	12.5	10.1	8.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	11.3	11.6	13.5	16.9	18.3	20.0	19.3	18.2	15.6	13.5	12.3
		MCDB	11.8	12.1	13.5	16.4	21.0	22.6	23.9	22.5	19.6	16.1	13.9	12.8
	2%	WB	10.1	10.2	10.5	12.2	15.1	16.9	18.5	17.7	16.6	14.4	12.3	11.1
		MCDB	10.7	11.0	11.9	14.8	18.4	20.2	21.6	20.0	18.0	15.1	12.8	11.7
	5%	WB	9.2	9.2	9.6	10.9	13.8	15.6	17.2	16.9	15.5	13.6	11.2	10.1
		MCDB	9.8	10.0	10.9	13.1	16.6	18.4	19.7	18.7	16.9	14.4	11.8	10.7
	10%	WB	8.1	8.2	8.7	10.0	12.5	14.6	16.2	16.0	14.6	12.7	10.2	8.9
		MCDB	8.9	9.0	10.0	12.1	14.9	17.0	18.4	17.7	15.9	13.6	10.9	9.6
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	5.3	6.1	6.9	7.6	7.3	6.9	6.8	6.4	5.6	4.8	4.7
		MCDBR	5.2	5.6	6.8	8.5	10.0	10.1	9.4	8.4	7.3	5.7	5.0	5.6
	5% WB	MCWBR	5.0	5.0	4.9	5.4	6.4	6.1	5.7	5.2	4.9	4.6	4.7	5.4
		MCWBR	5.3	5.5	6.4	7.5	9.3	9.1	8.5	7.4	6.5	5.4	4.9	5.7
(40) Clear-Sky Solar Irradiance	taub	taub	0.337	0.347	0.370	0.383	0.388	0.379	0.381	0.365	0.360	0.345	0.330	0.325
		taud	2.215	2.296	2.277	2.281	2.313	2.371	2.369	2.428	2.439	2.407	2.280	2.183
	Ebn at Noon	Ebn at Noon	570	722	794	839	856	867	857	851	805	722	579	487
		Edh at Noon	62	80	102	116	119	113	112	99	86	71	57	52
(44) All-Sky Solar Radiation	RadAvg	0.56	1.24	2.29	3.73	4.76	4.84	4.46	3.63	2.57	1.46	0.69	0.41	0.41
	RadStd	0.06	0.11	0.21	0.33	0.43	0.48	0.35	0.30	0.14	0.13	0.07	0.04	0.04

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page